PRESIDENT'S OFFICE

REGIONAL ADMINISTRATION AND LOCAL GOVERNMENT

MOROGORO MUNICIPAL COUNCIL

FORM THREE TERMINAL JOINT EXAMINATION

CODE: 031 PHYSICS

TIME: 3HRS MAY, 2025

INSTRUCTIONS:

- This paper consists of sections A, B and C with total of **eleven (11)** questions
- Answer all questions in sections A and B and **two (2) questions from section C**
- Non- programmable calculators and mathematical tables may be used

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SECTION A (16 Marks)

Answer **all** questions in this section

1. For each of the items (i) –(x), choose the correct answer from the given alternatives and write its letter beside the item number in the answer sheet provided.

i. A hospital technician uses X-ray machines to examine broken bones. Which best describes the role of physics in this situation?

A. Physics helps in designing comfortable beds

B. Physics explains how chemicals react inside the body

C. Physics allows imaging using electromagnetic radiation

D. Physics calculates the patient's body temperature

ii. A tailor uses a measuring tape to take a customer's waist size. Which physical quantity is being measured and why is accuracy important in this context?

A. Volume – to measure clothes' tightness

B. Length – to ensure a perfect fit

C. Area – to design clothing patterns

D. Mass – to determine cloth weight

iii. A sample of milk diluted with water has density of 1040kg/m3. if pure milk has a density of 1080kg/m3, then the percentage of water by volume in the diluted milk is

a. 55
b. 40
c. 50
d. 45

iv. A girl walks on soft ground wearing high-heeled shoes and leaves deep marks. Which physics concept explains this observation?

A. Force is spread over a large area

B. The ground is too weak

C. Pressure increases with small surface area

D. Gravity is weaker under her feet

v. A student walks on a woolen carpet with plastic shoes and receives a shock when touching a metal door handle. This is because:
A. The carpet conducts electricity
B. The metal handle is hot
C. The shoes generate kinetic energy
D. Static charges built up on the body discharged to the handle

vi. A uniform beam is supported at both ends and carries a weight in the middle. If the beam is in equilibrium, what is true about the reactions at the supports?
A. They are both zero
B. One is twice the other
C. They are equal
D. They act downward

vii. Why does a coin placed on a card fall into a glass when the card is quickly flicked away horizontally?
A. The force of the flick pulls the coin down.
B. The glass attracts the coin.
C. The coin remains at rest due to inertia.
D. Air pushes the coin downward.

viii. Which of the following best illustrates the use of the parallelogram law in real life?
A. A car braking on a flat road will experience minimum force compared to the braking in slanted surface
B. A person jumping vertically
C. Two people pulling a stuck object with ropes at an angle
D. A ball falling straight down

ix. In real life situation it more difficult to push a loaded wheelbarrow on a muddy surface than on a paved road. What do you think the reason behind?
A. Mud is heavier than the wheelbarrow
B. There is more air resistance on a muddy road

C. The wheelbarrow becomes heavier on mud

D. Frictional force is higher due to the irregular surface

x. When a ray of light passes from air into glass, it bends towards the normal. What does this tell us about the speed of light in glass compared to air?

A. Light travels faster in glass than in air

B. Light travels at the same speed in both media

C. Light is absorbed completely in glass

D. Light slows down in glass

2. Match each item in **Column A** with the correct description or principle from **Column B** by writing the letter of the correct response beside the question number.

List A	List B
i) Microscope ii) Periscope iii) Telescope iv) Camera v) Projector vi) Human eye	A. Uses two convex lenses to magnify very small objects for detailed observation B. Forms an inverted real image on a light-sensitive surface using a convex lens C. Allows viewing over obstacles using the principle of reflection in plane mirrors D. Produces a large virtual image by projecting light through a convex lens onto a screen E. Has a lens system and retina that works like a natural camera F. Uses a large objective lens to collect light from distant objects and magnify them G. It uses more than five lenses H. Both virtual and real images can be formed

SECTION B (54 Marks)

Answer **all** questions in this section.

3. a) During a practical session, your classmate accidentally spills a strong acid on the bench. What immediate steps should be taken to deal with the spill**?** **(4 Marks)**

   b) Rocket moves from the earth to a planet **x**. if it weighs 10, 000N and 300N on the earth and on planet **x** respectively determine the acceleration due to gravity on planet x (use $= 10N/kg$) **(5 Marks)**

4. A bottle of perfume is opened at one corner of a room, and soon the scent is smelled on the other side.

   a) (i) What property of matter is being demonstrated, **(2Marks)**

   (ii) Explain how this property allows the smell to spread across the room**. (2 Marks)**

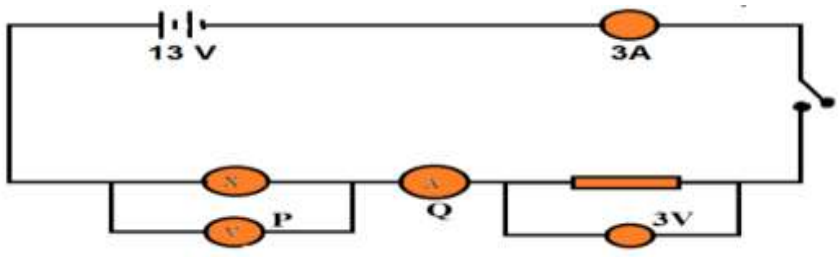
   b) A Small quantity of powdered charcoal is placed in a 100cm^3 density bottle and the total mass is measured to be 255g. If the mass of empty density bottle is 20g and when the battle is filled with water the total mass is 289g, what is the density of the powdered charcoal? **(5 Marks)**

5. a) You enter a dark room and switch on a torch. You immediately see objects in front of you,

   i) what property of light allows you to see those objects? **(2 Marks)**

   ii) Explain how the light helps you to see the objects. **(2 Marks)**

   b) Calculate the reading of the voltmeter, P and the ammeter, Q in the electric circuit battery



(5Marks)

6. a) Two bar magnets are suspended freely. When brought close, one rotates and aligns in the opposite direction to the other,

   i) What causes this rotation and alignment, **(2 Marks)**

   ii) State the law governing the phenomena about the magnetic poles of the magnets? **(2 Marks)**

b) A simple pulley system has velocity ratio of 3. Its efficiency is 90%, what load can it raise by an effort of 100N. **(5 Marks)**

7. a) A thermometer reads 37°C when placed under the armpit of a patient.
   i) What does this reading indicate? **(2Marks)**
   ii) Why is it important to measure body temperature in hospitals? **(2Marks)**

   b) A car moves 8km due to South and then suddenly changes its direction and move another 6km due West. Determine

   i) Total distance covered and displacement of the car. **(2.5 Marks)**

   ii) Direction of the displacement **(2.5 Marks)**

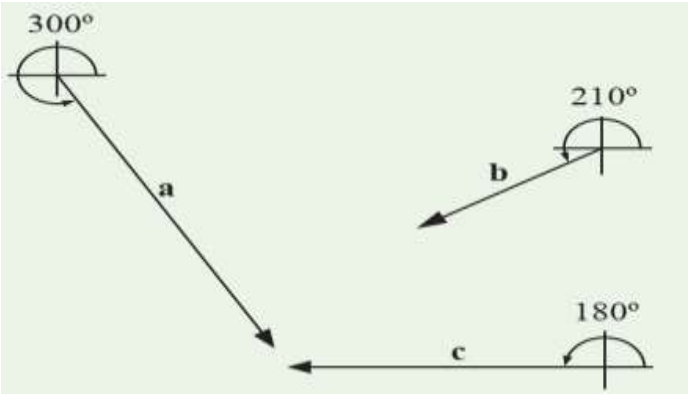
8. a) A rural village uses firewood for cooking, leading to deforestation. Suggest an alternative sustainable energy source and explain two benefits it would bring to the community. **(4Marks)**

   b) A block of concrete 5.0 m long expands to 5.00412 m when heated from 25 $C^0$to 100 C^0. Determine the linear expansivity of concrete **(5 Marks)**

SECTION C (30 Marks)

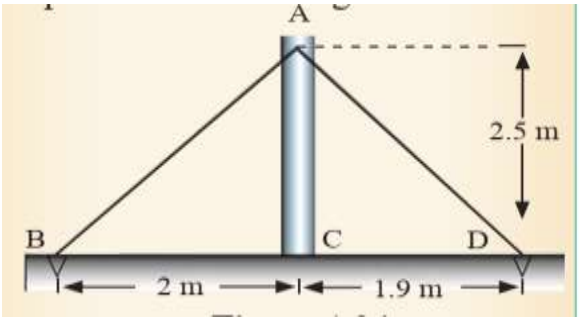
Answer **two (2**) questions from this section**.**

9. a) Figure below is a vectors representation given **a** , **b** and **c** having magnitudes of 32 m, 16 m and 24 m, respectively. Use graphical method of vector addition to find resultant vector. **(5 Marks)**



b) A boat is moving at 8.9 km/h relative to the water in a river. Initially the boat at the bank aims straight to the opposite bank of the river which is 120.9 m wide. If the speed of the water in the river is 2.9 km / h, how far downstream will the boat be when it reaches the opposite bank? **(5 Marks)**

c) Two cables are used to support a pole as shown in figure below If the tension in cables AB and AD is 500N and 160N respectively, determine the magnitude and direction of the resultant force exerted by cables AB and AD at point A using the parallelogram law **(5 Marks)**



10. a) A block of mass *m* slides down a plane inclined at an angle of 60^0 to the horizontal with an acceleration equal to half of the acceleration due to gravity. Show that the coefficient of kinetic friction, $\mu k = \sqrt{3} - 1$ **(5 Marks)**

b) Starting from rest, the time taken by a body to slide down a 45^0 inclined plane with friction is twice the time taken to slide down the same distance in the absence of friction. Determine the coefficient of friction between the body and the inclined plane. **(5 Marks)**

c) Explain the following observation

i) Most of the vehicles have their engines directly over the drive wheels **(2.5 Marks)**

ii) It is easy to slip when there is soap water on the floor **(2.5 Marks)**

11. a) Using a ray diagram, show how colors of light bend at different angles when passing through a glass prism. **(5 Marks)**

b) Find the index of refraction for a certain medium assuming light in air enters the medium at an incident angle of 30^0 and angle of refraction is 22^0 and sugest the type of medium in which light entered. **(5 Marks)**

c) A cube of sides 2cm is completely submerged in water so that the bottom of the cube is at a depth of 10cm. If the density of water is $1027 kg/m^3$ and g=10N/kg. What is the difference between the pressure at the bottom of the cube and pressure at the top? **(5 Marks)**